

Nazmus Shakib Shadin

Curriculum Vitae

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Education

- Fall 2024– Present **Doctor of Philosophy (Ph.D) in Computer Science**, *Kennesaw State University*, Marietta, Georgia, United States of America,
Ongoing GPA: 4.00 out of 4.00
- 2017–2022 **Bachelor of Science (B.Sc.) in Computer Science and Engineering**, *Ahsanullah University of Science and Technology*, Dhaka, Bangladesh,
CGPA: 3.483 out of 4.00

Publications

- 2026 **Nazmus Shakib Shadin**, Xinyue Zhang, Jingyi Wang, and Miao Pan. QSplitFL: Capability Aware Deep Q-Learning for Optimal Split Point Selection in Split Federated Learning, Accepted in 2026 European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD) as a Main Track Paper. Acceptance Rate < 24%.
- 2026 **Nazmus Shakib Shadin**, Aaron Cummings, Xinyue Zhang, and Bobin Deng. FedMTFI: Feature Importance Based Optimized Multi Teacher Knowledge Distillation in Heterogeneous Federated Learning Environment, Accepted in 2026 International Joint Conference on Neural Network (IJCNN) as a Main Track Paper.
- 2026 **Nazmus Shakib Shadin**, Xinyue Zhang, and Dan Chia-Tien Lo. Kernel-Aware Federated Learning with Multi-Teacher Knowledge Distillation and OS-Level Synchronization, **10.1145/3746467.3801500**. Published in 2026 The ACM Southeast (ACMSE) conference as a Main Track Full Paper.
- 2026 Md Mushfique Hossain, **Nazmus Shakib Shadin**, Aaron Cummings, Xinyue Zhang, and Modupe Akintomide. DualCareKG: Integrating Clinical and Patient Knowledge Graphs for Dementia Caregiver Support, Accepted in IEEE/ACM conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE).
- 2026 Mohammad Yusufi, **Nazmus Shakib Shadin**, Md Mushfique Hossain, Xinyue Zhang, and Rifatul Islam. How Much Does Privacy Cost? Evaluating Differential Privacy for ML based VR Gaze and Head Tracking, Accepted in 2026 IEEE Computers, Software, and Applications Conference (IEEE COMPSAC).
- 2026 Nasim Ahmed, Md Mahedi Hassan, Md Mushfique Hossain, **Nazmus Shakib Shadin**, Xinyue Zhang, and Rifatul Islam. Obscuring the ‘Who,’ Preserving the ‘What’: Targeted Eye-tracking Feature Obfuscation in Virtual Reality for Privacy-Utility Balance, **10.1109/VR67842.2026.00047**. Published in 2026 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) as a Main Track Paper.
- 2025 **Nazmus Shakib Shadin**, Xinyue Zhang, Jingyi Wang, and Miao Pan. Heterogeneity-Aware Private Personalized Federated Learning for Medical Imaging via Contrastive Distillation, **10.1109/BigData66926.2025.11401860**. Published in 2025 IEEE International Conference on Big Data (BigData) as Regular Main Track Paper (Acceptance rate: 18.7%).

- 2025 **Nazmus Shakib Shadin**, Nasim Ahmed, Md Mahedi Hassan, Rifatul Islam, and Xinyue Zhang. YouKnowWho: Interpretable Framework for Classifying the Gender from Behavioral Motion Data in Extended Reality (XR), **10.1145/3704413.3765309**. Published in The Twenty-sixth ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc '25), Houston, TX, USA.
- 2025 **Nazmus Shakib Shadin** and Xinyue Zhang. FedKDSHap: Enhancing Federated Learning via Shapley Values Driven Knowledge Distillation on Non-IID Data, **10.1145/3701716.3717645**. Published in Companion Proceedings of the ACM Web Conference 2025 (WWW Companion '25) FL@FM workshop, Sydney, NSW, Australia.
- 2022 **Nazmus Shakib Shadin**, Silvia Sanjana, Nusrat Jahan Lisa. Covid-19 Detection from Lung CT Scan using Transfer Learning Models, **10.1007/978-3-031-17181-9_5**. Published in Springer Communications in Computer and Information Science Series (CCIS, volume 1550).
- 2022 **Nazmus Shakib Shadin**, Silvia Sanjana, and Daiyan Ibrahim. Face Mask Detection using Deep Learning and Transfer Learning Models, **10.1109/ICISSET54810.2022.9775925**. 2022 International Conference on Innovations in Science, Engineering and Technology (ICISSET).
- 2022 **Nazmus Shakib Shadin**, Silvia Sanjana, Sovon Chakraborty, and Nusrat Sharmin. Performance Analysis of Glaucoma Detection using Deep Learning Models, **10.1109/ICISSET54810.2022.9775828**. 2022 International Conference on Innovations in Science, Engineering and Technology (ICISSET).
- 2021 **Nazmus Shakib Shadin**, Silvia Sanjana, and Nusrat Jahan Lisa. Covid-19 diagnosis from chest x-ray images using convolutional neural network(cnn) and inceptionv3, **10.1109/icit52682.2021.9491752**. In 2021 International Conference on Information Technology (ICIT), pages 799–804, 2021.
- 2021 FM Javed Mehedi Shamrat, Sovon Chakraborty, Md Masum Billah, Moumita Kabir, **Nazmus Shakib Shadin**, and Silvia Sanjana."Bangla numerical sign language recognition using convolutional neural networks", **10.11591/ijeecs.v23.i1.pp405-413**. Indonesian Journal of Electrical Engineering and Computer Science, 2021.
- 2021 **Nazmus Shakib Shadin**, Silvia Sanjana, and Mayisha Farzana. Automated detection of covid-19 pneumonia and non-covid-19 pneumonia from chest x-ray images using convolutional neural network (cnn), **10.1109/icitech50181.2021.9590174**. In 2021 2nd International Conference on Innovative and Creative Information Technology (ICITech), pages 57–63, 2021.
- 2021 Silvia Sanjana, **Nazmus Shakib Shadin**, and Mayisha Farzana. Automated Diabetic Retinopathy Detection using Transfer Learning Models, **10.1109/ICEEICT53905.2021.9667793**. 2021 5th International Conference on Electrical Engineering and Information Communication Technology (ICEEICT).

Work Experience

- August 2024 to Present **Graduate Research Assistant, AIPO Lab, Kennesaw State University, Marietta, GA, USA**
- Optimize decentralized learning environments with heterogeneous data and models.
 - Develop energy-efficient strategies for heterogeneous Federated Learning (FL).
 - Apply knowledge distillation (KD) to improve computational efficiency in FL.
 - Investigate and Experiment techniques for reducing model size while maintaining performance metrics.
 - Utilize feature importance to optimized KD Loss function using Shapley Values.
 - Develop techniques ensuring secure and efficient privacy-preserving AI systems.

- August 2022 to July 2024 **Software Development Engineer-I, IQVIA, Dhaka, Bangladesh**
- Part of the PADDs (Processing and Data Delivery System) Implementation Team at IQVIA
 - Developed and optimized data-driven solutions for pharmaceutical and healthcare analytics.
 - Implemented data processing workflows and predictive analytics using Python, PowerBuilder.
 - Developed machine learning, deep learning, and transfer learning techniques for predictive analytics.
 - Contributed to full-stack development for PADDs using ASP.NET, TypeScript, and JavaScript.
 - Integrated web services for seamless data communication.
 - Worked with SQL, Sybase, and Exasol for large-scale database management.
 - Optimized data retrieval and processing for high-performance applications.
 - Worked with cloud platforms such as Azure, AzureML, Azure DevOps etc.
- January 2022 to August 2022 **Software Developer Intern, IQVIA, Dhaka, Bangladesh**
- Gained hands-on experience in designing and optimizing data-driven solutions.
 - Learned data warehousing techniques and business intelligence (BI) tools for decision-making.
 - Worked with SQL, focusing on data modeling and performance tuning.
 - Developed automation scripts to streamline workflows and improve system efficiency.
 - Gained practical experience with SSIS (SQL Server Integration Services) for ETL processes.
 - Applied project management methodologies to contribute effectively to real-world applications.

Technical Skills

- Languages **Python, C, C++, Java, JavaScript, TypeScript, SQL, PL/SQL, C#**
- Cloud **Azure, Azure ML, Apache Spark, Azure DevOps, Databricks**
- Database **PostgreSQL, MySQL, Exasol, Sybase, SQL Server**
- Frameworks **ASP.NET, ASP.NET Core, VBA, PowerBuilder, PowerScript, TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, YOLOv8, GraphViz, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Git, SSIS**

Undergraduate Research

Ahsanullah University of Science and Technology, Dhaka, Bangladesh

Title: **A Comparative Study of Diabetic Retinopathy Detection using Machine Learning and Deep Learning architectures.**

Used six deep learning and machine learning models to detect the Diabetic Retinopathy efficiently with the retinal fundus image using digital image processing. **Github Repository**

Supervisor: **Dr.-Ing. Nusrat Jahan Lisa**, *Assistant Professor, Department of Computer Science and Engineering*, Ahsanullah University of Science and Technology

Projects

- 2021 **Sentiment Classification from Tweeter Data using Bi-directional LSTM.** NLP based project that applies to classify sentiment of a person using the Tweeter based data using Bi-directional LSTM. **Github Repository**
- 2021 **Door Step Pharmacy.** Distributed Database Management System based in "PL/SQL" Language, and Run in CLI. **Github Repository**
- 2020 **Online Disease Checker and Medical Support (ODCMS).** Developed In Microsoft Visual Studio 2019 using asp.net, C#, JavaScript, and CSS. **Github Repository**
- 2020 **Pharmacy Management System.** Developed In Microsoft SQL Server and NetBeans using SQL and JAVA. **Github Repository**
- 2020 **Learn-C.** An android app developed in Android Studio 4.0.1 using JAVA. **Github Repository**
- 2019 **Media Player.** A desktop app developed in NetBeans using JavaFx. **Github Repository**

Services

- 2025–2026 Program Committee Member and Reviewer, *IEEE International Conference on Tools with Artificial Intelligence* (ICTAI).
- 2025–2026 Reviewer, *Medical Image Computing and Computer Assisted Intervention* (MICCAI).
- 2025–2026 Reviewer, *International Conference on Computing, Networking and Communications* (ICNC): AI and Machine Learning for Communications and Networking.
- 2026 Reviewer, *International Joint Conference on Neural Networks* (IJCNN)
- 2026 Reviewer, *IEEE Internet of Things* (IoT) Journal.
- 2026 Reviewer, *International Conference on Intelligent Multimedia, Networking, and Security* (IMNS).
- 2025 Reviewer, *European Conference on Artificial Intelligence* (ECAI).
- 2025 Reviewer, *ACM Secure and Trustworthy Cyber-Physical Systems* (SaT-CPS 2025).
- 2025 Reviewer, *Generative AI for Smart and Connected Health: Innovations, Challenges, and Applications* (GenAI4SCH).
- 2025- Association for Computing Machinery (ACM) Student Member (*2025-present*)
- 2026- IEEE Student Member and IEEE CIS member (*2026-present*)
- 2026- International Neural Network Society (INNS) Student Member (*2026-present*)

Professional Certification

- DP-100: *Microsoft Certified: Azure Data Scientist Associate*. **Credential Badge**
- AI-900: *Microsoft Certified: Azure AI Fundamentals*. **Credential Badge**
- AZ-900: *Microsoft Certified: Azure Fundamentals*. **Credential Badge**

Honors and awards

- Dean' 4.0 *Kennesaw State University - (CCSE) College of Computing and Software Engineering* **Got 4.0 Club: out of 4.0 and listed in Dean's 4.0 Club.** Awarded on December 2024. Credential ID: 2024-2026 a1BRH6CrQ1G8bMhusSiyTA. **Credential Badge**
- Scholarship *Associated with Ahsanullah University of Science and Technology.* **Got a 50 percent Tuition - Spring Scholarship from Ahsanullah University Of Science And Technology for attaining 2020 a place among the top 5 percent of students.**

References

Dr. Xinyue Zhang

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Department of Computer Science
Kennesaw Sate University,
USA*
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Dr. Nusrat Jahan Lisa

*Assistant Professor,
Department of ECE
North South University,
Bangladesh*
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